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Post-Operative Rehabilitation and Return to Sport Criteria Following Distal Biceps Tendon Rupture Surgery

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ABSTRACT

BACKGROUND: Within the clinical literature, there is no consensus on optimal rehabilitation for return to sport (RTS) after distal biceps tendon rupture (DBTR) repair. The authors hypothesize that surgeons will utilize time-based criteria for RTS rather than performance-based or functional criteria for RTS.

METHODS: Level 1 to 4 studies that evaluated DBTR with a minimum of 12 months follow-up were eligible. Exclusion criteria included studies that treated DBTR nonoperatively, lacked RTS criteria, or included revision surgery. Studies were evaluated for RTS criteria, timeline/rate of RTS, and methodologic quality.

RESULTS: Out of 671 studies identified, 5 met inclusion criteria. These included 268 patients playing 22 different sports. Timeline for return ranged from 3 to 130 weeks. Range of motion exercises followed by strengthening (100%) and multiphase rehabilitation program (100%) were the most common parameters emphasized in rehabilitation protocols, followed by immediate postoperative cast/splint immobilization (80%). All studies included a timeline for RTS; however, no article listed subjective or specific objective measurement criteria for RTS.

CONCLUSION: There is no consensus on RTS criteria in the literature. Eighty percent of the included studies reported utilizing initial cast/splint immobilization, 100% reported a multiphase program consisting of some form of progressive ROM followed by strengthening. Sixty percent reported a supervised physical therapy routine. One hundred percent reported timing after surgery as a criterion for return to sport, but this timing varied greatly across the included studies.

LEVEL OF EVIDENCE: Level IV; Systematic Review

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KEYWORDS: distal biceps tendon rupture, distal biceps repair, return-to-sports criteria, rehabilitation, RTS protocol for DBTR, functional criteria for distal biceps repair

The biceps brachii muscle aids in supination strength and contributes to elbow flexion.[24] Distal biceps tendon rupture (DBTR) most commonly results from an eccentric force on a contracted muscle belly. Incidence of distal biceps tendon rupture in the general population is about 5/100,000.[3] This usually occurs in males in their mid-30s through mid-50s, but it also occurs in younger athletes, particularly weight lifters.[9] Early surgical repair is recommended for nearly all patients due to severe limitations in flexion and supination strength with non-operative management. Repair of the distal biceps is especially important for athletes in regards to return to sport (RTS).[6]

Distal biceps tendon rupture is relatively common, yet no set criteria exist for functional measures, including elbow range of motion or flexion/supination strength, which would determine whether a player is ready to RTS following surgery for DBTR. Furthermore, there is no universally accepted rehabilitation program for RTS following surgery. Various programs have been reported, but a universal protocol has not been adopted.[12, 18]

The purpose of this study was to perform a systematic review of the current literature and assess postoperative rehabilitation parameters as well as RTS criteria reported in distal biceps tendon rupture scientific literature. The authors hypothesized that the majority of studies would utilize primarily time-based criteria and not performance based or objective functional data to determine when to allow athletes to return to sport.

MATERIALS AND METHODS:

Search Strategy

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A systematic review of level 1 to 4 studies was performed to evaluate patients who underwent operative intervention of distal biceps rupture. Literature search was conducted using PubMed, CINAHL, MEDLINE, SPORTDiscus, and Cochrane databases from inception to March 2022. Search criteria were a combination and variation of biceps tendon, distal biceps rupture, surgery, reconstruction, outcomes, rehabilitation, and return to play. Once studies that met inclusion criteria were selected, an additional search of all references was performed to find any additional potential articles that were missed. The review was developed and reported in accordance with PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines and registered in PROSPERO (Prospective Register of Systematic Reviews).

Initial search and review were performed by two blinded independent reviewers. After unblinding, any discrepancies in study eligibility between the two reviewers was reviewed by an additional third reviewer. In all cases a consensus was reached. Review was completed under direct supervision of the senior author. The studies that met inclusion criteria were compiled and extracted for data (Figure 1).

Eligibility Criteria

Inclusion criteria included studies conducted on patients who underwent operative intervention of distal biceps rupture, had at least one year of postoperative follow-up, and were written in English. Exclusion criteria included studies that treated distal biceps rupture nonoperatively, had less than one year follow-up, lacked explicit return to sport or rehabilitation criteria, or included revision procedures. Review articles, case series, cadaveric or biomechanical studies, and abstracts were also excluded. If an article included data from a previously published patient group or had overlapping study participants, the most recent article was included in this study.

Data Extraction and Statistical Analysis

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Data extracted included number of patients, demographics, hand dominance, sport played, injury setting, level of play/competition, surgical technique, rehabilitation protocol, return to sport time, follow-up duration, and complications. Patient reported outcomes (PROs) were not reported due to variety of surveys used. All data were collected and analyzed using Microsoft Excel.

Quality of Literature Methodology

Methodological quality of included studies was independently assessed using a modified Coleman Methodology score (MCMS) by two reviewers (BP, AB).[15] The Coleman Methodology Score (CMS) assesses study methodology based on 10 criteria and has a final score ranging from 0 to 100 with scores from 85 to 100 considered excellent, 70 to 84 good, 55 to 69 fair, and less than 55 poor quality. The subsections included in the CMS are based on the subsections of the Consolidated Standards of Reporting Trials (CONSORT) statement and can be modified depending on the included trial design.[8] Regarding RTS criteria, each article was scored from 0 to 4 according to methodology presented by Zaman et al.[26]

RESULTS:

Six hundred and seventy-one published articles were found after initial search and exclusion of duplicates. Upon implementation of inclusion and exclusion criteria, 5 studies were included in the final analysis.[4, 7, 14, 17, 25] Two studies included patients that all identified as active in sports (Gowd et al, Yetter et al), and the other three included both athletes and non-athletes. Since there was no separation within the data for these groups, the entire population of each study was included. There was a total of 268 patients, 206 male, 1 female, and 61 for whom sex data was unavailable. While most studies did not report what sports were played, there were injuries across multiple sports (Table 1). Level of play was only reported in one study (Rantanen). Two studies reported on if patients were able to return to the same or different level of play

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postoperatively. Three studies were level 3 and two studies were level 4. Both single and dual incision techniques were used with multiple fixation strategies including suture passed through bone tunnels, suture anchors, and cortical button.

Quality of Literature

The quality of the literature according to the modified Coleman Methodology Score was 61.6 ± 6.66 which is considered fair (Table 2). All studies reported good certainty of the diagnosis preoperatively, description of the surgical technique, description of the postoperative rehabilitation, and adequate follow-up.

Parameters of Postoperative Rehabilitation

All 5 studies detailed at least some elements of the postoperative rehabilitation parameters. Details of the protocols are summarized in Table 3. Five main components of the rehabilitation protocols were examined. All studies had some amount of range of motion exercises with progression to strengthening, although the amount of time spent in each phase varied between the studies. Four studies (80%) used some period of cast or splint immobilization ranging from 2 to 5 weeks. Yetter et al did not use a cast or splint immobilization post operatively, but only a bulky soft dressing for 6 weeks.1 Carter et al used a splint for 76% of patients while 24% were initially placed in a bulky dressing and sling, which was only mentioned as surgeon preference and was not based on any preoperative demographics or intra-operative findings.[4] No studies indicated sport specific rehab protocols, although Gowd et al did mention that sport specific exercises were advanced as tolerated after 12 weeks post op.[7] Only three studies (60%) used a supervised physical therapy program.

RTS Criteria

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Time from surgery. The timelines for RTS are listed in Table 4. Only Gowd et al and Carter et al detailed the RTS timeline. All studies had a recommended RTS time in their rehabilitation protocol, but three studies (60%) did not report timing of return to sport. McKee et al did report that compliance with return to aggressive activity was variable and that many patients returned to full activity before the recommended timeline.

Conditional RTS Criteria. The articles were analyzed for any criteria, whether subjective or objective, that were utilized as a barrier to sport and continued therapy. No studies reported any such criteria. Only Rantanen et al detailed a threshold for progressing from ROM exercises to weight training, requiring a near normal elbow range of motion.[17]

Specific Measurements for RTS. None of the studies reported the use of specific clinical measurements as criteria, including ROM or strength requirements. Additionally, there were no imaging assessments utilized.

Assessment of Value. Each study was assessed for value based on prior methodology presented by Zaman et al (Table 5).[26] This methodology assigns a score for 4 categories: rehabilitation protocol, timeline for return to activity, objective/subjective criteria, and specific measurement criteria for a possible total of 4. All studies reported the same two metrics: rehabilitation protocol and timeline for return to activity. No studies reported objective/subjective criteria or specific measurement criteria. Similar to Zaman et al original review, no studies reached 4/4 on the scale. This means no study completely defined return to sport criteria.

Rate of RTS. RTS rate was reported in 4 of the 5 studies. One study (Carter et al) reported overall RTS rate with 51 of 62 (82.3%) returning to some level of sport.[4] Yetter et al reported RTS as those with or without difficulty returning to sport but not specifically if they were able to

return.[25] The overall return without difficulty was 70%. Within this study, the difficulty with return to sport was further broken down by mental component score (MCS) with 89% of those with a MCS < 50 reporting difficulty returning while only 8% of those with MCS > 50 reporting difficulty returning to sport. Gowd et al and Rantanen et al both reported RTS rate and compared the pre and postoperative level of sport. Rantanen et al reported that 14/15 (93%) of patients returned to the same level of sport. 15/15 (100%) returned to some level of sport.[17] Gowd et al reported that 57/61 (93%) returned to some level of sport while 40/61 (65.6%) returned to sport at the same or higher intensity. McKee et al reported that 13/53 (25%) of patients' initial injury was during playing a sport but did not comment on how many total athletes were in the study population or how many ultimately returned to sport. They do report that 81% of patients were satisfied with their outcome. The overall RTS rate among the 4 studies in which it was reported was 146/171 (85%)

DISCUSSION:

The findings of this study demonstrate the need for a validated rehabilitation and return to sport protocol. While all of the studies included multiphase progressive ROM and strengthening programs, the primary criteria utilized for return to sport was time-based. Within this time-based criterion, there was a large range from 8 weeks to 6 months.

Progressive range of motion parameters have been a staple of postoperative rehab for tendon repair for many years. Historically, a period of immobilization has been used prior to performing ROM exercises in tendon repairs. This period typically lasted anywhere from 2 to 6 weeks. A recent randomized control trial by Bergman et al evaluated the need for immobilization for return to work in distal biceps repair.[2] This study used a single incision, cortical button technique and found no significant difference in return to work between a control group

immobilized for 6 weeks and the treatment group who were self-weaned from a sling with active ROM as tolerated during the first 6 weeks. While this study showed no significant difference in postoperative ROM between the groups, there was a significant decrease in passive forearm supination (p=0.04) in the control group.

Although the return to sport protocols and criteria are lacking in the current literature, there have been numerous recent studies investigating the strongest and most biomechanically sound surgical fixation for distal biceps tendon rupture.[1, 11, 20-23] There have also been analysis of rates of return to work. Rubinger et al performed a systematic review for return to work following distal biceps repair. They found 40 studies that met inclusion criteria with a total of 1270 patients. 89% of patients were able to fully return to work and the average time to return to work was 14.37 weeks.[19] This return to function percentage is similar to our findings in RTS studies, but the average time to return to work was less than the time to return to sport found in the included studies.

Many professional sports teams, team physicians, and training personnel including athletics trainers and physical therapists likely have more nuanced sport specific return to sport rehabilitation protocols than the ones currently published. However, the paucity of published protocols and specific RTS criteria remains a limitation. While there was a high rate of return to play in the included studies, there is a lower return to play at the same or higher level after injury. Gowd et al reported only a 65.6% return to sport at same or higher intensity.[7] Rantanen et al reported a 93% RTS rate, but in patients who were involved in competitive lifting or competitive overhead sport, the return to sport rate at the same level was only 86%.[17] There were two recent studies that reviewed RTS in the NFL after distal biceps surgical repair.[13, 16] Pagani et al examined data for players who underwent distal biceps tendon repair during a 20 year period

from 1998-2017 and found 84% were able to return to play for at least one game. McGinniss et al identified 35 NFL players from 2000-2016 with a RTS rate of 94%. While both of these studies demonstrate a high return to sport rate in high level athletes, as they are retrospective reviews of publicly available data, they cannot comment on rehabilitation protocols or specific return to sport criteria that could be potentially be applied across other sports or levels of play. Further studies are needed to investigate sport specific protocols and if there is a need for variable protocols for competitive vs recreational return to play.

Limitations

The primary limitation of this study is the small sample size and reliance on retrospective data. Multiple studies surveyed patients at time points well past the index procedure. Due to the multiple patient reported outcome scoring systems used, the decision was made not to include these patient reported outcomes. Additionally, the level of sport was not reported in most of the included studies. Recreational athletes likely have a more standardized return to play protocol while professional athletes have more sport specific protocols. Various physicians may also have published rehabilitation protocols for patients and therapists but have not published these protocols in peer reviewed literature. Our review followed the methodology used in previous studies of return to sport criteria and excluded these published protocols.[5, 10, 26] Finally, there were multiple surgical techniques utilized in these studies that ranged over 20 years, which could increase the variability in postoperative protocols and ultimate return to sport.

CONCLUSIONS:

There is no consensus on return to sport criteria in the literature. Eighty percent of the included studies reported utilizing initial cast/splint immobilization and 100% reported a multiphase program consisting of some form of progressive ROM followed by strengthening. Sixty percent

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reported a supervised physical therapy routine. One hundred percent reported timing after surgery as a criterion for return to sport, but this timing varied greatly across the included studies. Future studies should focus on a checklist of subjective and objective performance-based criteria for safe return to sport.

REFERENCES:

1. Amini MH. Two-Incision Distal Biceps Repair with Cortical Button: A Technique to Improve Supination Strength. Arthrosc Tech 2021;10(7):e1859-e63. doi:10.1016/j.eats.2021.04.006
2. Bergman JW, Silveira A, Chan R, Lapner M, Hildebrand KA, Le ILD, et al. Is Immobilization Necessary for Early Return to Work Following Distal Biceps Repair Using a Cortical Button Technique?: A Randomized Controlled Trial. J Bone Joint Surg Am 2021;103(19):1763-71. doi:10.2106/JBJS.20.02047
3. Carrazana-Suarez LF, Cooke S, Schmidt CC. Return to Play After Distal Biceps Tendon Repair. Curr Rev Musculoskelet Med 2022;15(2):65-74. doi:10.1007/s12178-022-09742-x
4. Carter TH, Karunaratne BJ, Oliver WM, Murray IR, White TO, Reid JT, et al. Acute distal biceps tendon repair using cortical button fixation results in excellent short- and long-term outcomes : a single-centre experience of 102 patients. Bone Joint J 2021;103-B(7):1284-91. doi:10.1302/0301-620X.103B7.BJJ-2020-2246.R1
5. Ciccotti MC, Syed U, Hoffman R, Abboud JA, Ciccotti MG, Freedman KB. Return to Play Criteria Following Surgical Stabilization for Traumatic Anterior Shoulder Instability: A Systematic Review. Arthroscopy 2018;34(3):903-13. doi:10.1016/j.arthro.2017.08.293

Journal Pre-proof

6. Frank RM, Cotter EJ, Strauss EJ, Jazrawi LM, Romeo AA. Management of Biceps Tendon Pathology: From the Glenoid to the Radial Tuberosity. J Am Acad Orthop Surg 2018;26(4):e77-e89. doi:10.5435/JAAOS-D-17-00085

7. Gowd AK, Liu JN, Maheshwer B, Garcia GH, Beck EC, Cohen MS, et al. Return to sport and weightlifting analysis following distal biceps tendon repair. J Shoulder Elbow Surg 2021;30(9):2097-104. doi:10.1016/j.jse.2021.01.034

8. Jakobsen RB, Engebretsen L, Slauterbeck JR. An analysis of the quality of cartilage repair studies. J Bone Joint Surg Am 2005;87(10):2232-9. doi:10.2106/JBJS.D.02904

9. Kelly MP, Perkinson SG, Ablove RH, Tueting JL. Distal Biceps Tendon Ruptures: An Epidemiological Analysis Using a Large Population Database. Am J Sports Med 2015;43(8):2012-7. doi:10.1177/0363546515587738

10. Kemler BR, Rao S, Willier DP, 3rd, Jack RA, 2nd, Erickson BJ, Cohen SB, et al. Rehabilitation and Return to Sport Criteria Following Ulnar Collateral Ligament Reconstruction: A Systematic Review. Am J Sports Med 2022;50(11):3112-20. doi:10.1177/03635465211033994

11. Lang NW, Bukaty A, Sturz GD, Platzer P, Joestl J. Treatment of primary total distal biceps tendon rupture using cortical button, transosseus fixation and suture anchor: A single center experience. Orthop Traumatol Surg Res 2018;104(6):859-63. doi:10.1016/j.otsr.2018.05.013

12. Logan CA, Shahien A, Haber D, Foster Z, Farrington A, Provencher MT. Rehabilitation Following Distal Biceps Repair. Int J Sports Phys Ther 2019;14(2):308-17. No DOI

Journal Pre-proof

13. McGinniss A, Guinand LA, Ahmed I, Vosbikian M. Distal biceps ruptures in National Football League players: return to play and performance analysis. J Shoulder Elbow Surg 2021;30(7):1647-52. doi:10.1016/j.jse.2020.11.010

14. McKee MD, Hirji R, Schemitsch EH, Wild LM, Waddell JP. Patient-oriented functional outcome after repair of distal biceps tendon ruptures using a single-incision technique. J Shoulder Elbow Surg 2005;14(3):302-6. doi:10.1016/j.jse.2004.09.007

15. Nyland J, Causey B, Wera J, Krupp R, Tate D, Gupta A. Distal biceps brachii tendon repair: a systematic review of patient outcome determination using modified Coleman methodology score criteria. Knee Surg Sports Traumatol Arthrosc 2017;25(7):2293-7. doi:10.1007/s00167-015-3899-7

16. Pagani NR, Leibman MI, Guss MS. Return to play and performance after surgical repair of distal biceps tendon ruptures in National Football League athletes. J Shoulder Elbow Surg 2021;30(2):346-51. doi:10.1016/j.jse.2020.06.023

17. Rantanen J, Orava S. Rupture of the distal biceps tendon. A report of 19 patients treated with anatomic reinsertion, and a meta-analysis of 147 cases found in the literature. Am J Sports Med 1999;27(2):128-32.

18. Rollo G, Meccariello L, Rotini R, Pichierri P, Bisaccia M, Fortina M. Efficacy of the "Salento technique", a modified two-incision approach in distal biceps brachii tendon repair. Surgical description and outcomes analysis. J Clin Orthop Trauma 2019;10(5):959-64. doi:10.1016/j.jcot.2019.02.006

19. Rubinger L, Solow M, Johal H, Al-Asiri J. Return to work following a distal biceps repair: a systematic review of the literature. J Shoulder Elbow Surg 2020;29(5):1002-9. doi:10.1016/j.jse.2019.12.006

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20. Rutka V, Weppe F, Duprey S, Gras LL. Distal biceps tendon repair via new knotless endobutton fixation: A biomechanical study. Shoulder Elbow 2021;13(3):321-8. doi:10.1177/1758573219864303

21. Siebenlist S, Schmitt A, Imhoff AB, Lenich A, Sandmann GH, Braun KF, et al. Intramedullary Cortical Button Repair for Distal Biceps Tendon Rupture: A Single-Center Experience. J Hand Surg Am 2019;44(5):418 e1- e7. doi:10.1016/j.jhsa.2018.07.005

22. Tadevich JT, Bhagat ND, Lim BH, Gao J, Chen WW, Merrell GA. Power-Optimizing Repair for Distal Biceps Tendon Rupture: Stronger and Safer. J Hand Surg Glob Online 2021;3(5):266-71. doi:10.1016/j.jhsg.2021.06.004

23. van der Vis J, Janssen SJ, Haverlag R, van den Bekerom MPJ. Functional outcome in patients who underwent distal biceps tendon repair. Arch Orthop Trauma Surg 2018;138(11):1541-8. doi:10.1007/s00402-018-3018-6

24. von Werder SC, Disselhorst-Klug C. The role of biceps brachii and brachioradialis for the control of elbow flexion and extension movements. J Electromyogr Kinesiol 2016;28:67-75. doi:10.1016/j.jelekin.2016.03.004

25. Yetter T, Patton AG, Mansi A, Maassen N, Somerson JS. Low mental health scores are associated with worse patient-reported outcomes and difficulty with return to work and sport after distal biceps repair. JSES Int 2021;5(3):597-600. doi:10.1016/j.jseint.2020.12.015

26. Zaman S, White A, Shi WJ, Freedman KB, Dodson CC. Return-to-Play Guidelines After Medial Patellofemoral Ligament Surgery for Recurrent Patellar Instability: A Systematic Review. Am J Sports Med 2018;46(10):2530-9. doi:10.1177/0363546517713663

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FIGURES AND TABLES LEGENDS:

Figure 1: PRISMA (Preferred Reporting Items for Systematic Review and Meta-Analyses) flow diagram for the assessment strategy for included studies. RTS; return to sport

Table 1: Distribution of Sports in Total Patient Population

Table 2: Modified Coleman Methodology Scores with Means and SDs for Each Scoring Category[a]

[a] 1, study size (10): 2, percentage of patients with follow-up (10); 3, Number of interventions per group (10); 4, study type (15); 5, diagnostic certainty (5); 6, description of surgical technique (10); 7, description of postoperative rehabilitation (5); 8, outcome criteria (10); 9, procedure for assessing outcomes (15); 10, description of patient selection process (10). Total out of 100.

Table 3: Summary of Postoperative Rehabilitation Parameters

Table 4: RTS Timeline by Included Study

Table 5: Scoring Report for RTS Value Assessment[a]

[a] A maximum of 1 point was given per category if the study met the criteria. RTS, return to sport

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Table 1

Distribution of Sports in Total Patient Population

Sport Included	Number of Studies
Not Reported	3
Weight lifting/Body Builder	2
Arm Wrestling	1
Judo	1
Rugby	1
Wrestling	1
Hockey	1
Soccer	1
Baseball	1
Tennis	1
Volleyball	1
Football	1
Softball	1
Yoga	1
Downhill Skiing	1
Basketball	1
Crossfit	1
Bowling	1
Running	1
Swimming	1
Cycling	1
Fitness	1
Golf	1

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Table 1

Modified Coleman Methodology Scores with Means and SDs for Each Scoring Category[a]

Authorship	1	2	3	4	5	6	7	8	9	10	Total
Rantanen et al	0	10	0	0	5	10	5	7	8	5	50
Yetter et al	4	4	10	0	5	10	5	10	11	5	64
Gowd et al	7	7	7	0	5	10	5	7	11	5	64
McKee et al	7	4	10	0	5	10	5	10	11	5	67
Carter et al	10	7	0	0	5	10	5	10	11	5	63
Avg ± SD	5.6 ± 3.8	6.4 ± 2.5	5.4 ± 5.1	0 ± 0	5.0 ± 0	10.0 ± 0	5.0 ± 0	8.8 ± 1.6	10.4 ± 1.3	5.0 ± 0	61.6 ± 6.7

[a] 1, study size (10): 2, percentage of patients with follow-up (10); 3, Number of interventions per group (10); 4, study type (15); 5, diagnostic certainty (5); 6, description of surgical technique (10); 7, description of post-operative rehabilitation (5); 8, outcome criteria (10); 9, procedure for assessing outcomes (15); 10, description of patient selection process (10). Total out of 100.

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Table 3

Summary of Postoperative Rehabilitation Parameters

Postop Rehabilitation Parameters	Number of Studies
Cast/splint Immobilization	4
Range of motion exercises	5
Strengthening	5
Supervised Physical Therapy	3
Multiphase Program	5

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Table 4

RTS Timeline by Included Study

Authorship	Recommend RTS Timeline	RTS Timeline Reported
Rantanen et al	4-6 months	Not Reported
Yetter et al	6 months	Not Reported
Gowd et al	12 weeks	6 months ± 2.8
McKee et al	8-12 weeks	Not Reported
Carter et al	20 weeks	22.3 weeks (3 to 130)

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Table 5

Scoring Report for RTS Value Assessment[a]

Authorship	Rehabilitation Protocol	Timeline for Return to Activity	Objective/Subjective Criteria	Specific Measurement Criteria	RTS Article Rating Score (4)
Rantanen et al	1	1	0	0	2
Yetter et al	1	1	0	0	2
Gowd et al	1	1	0	0	2
McKee et al	1	1	0	0	2
Carter et al	1	1	0	0	2

[a] A maximum of 1 point was given per category if the study met the criteria. RTS, return to sport

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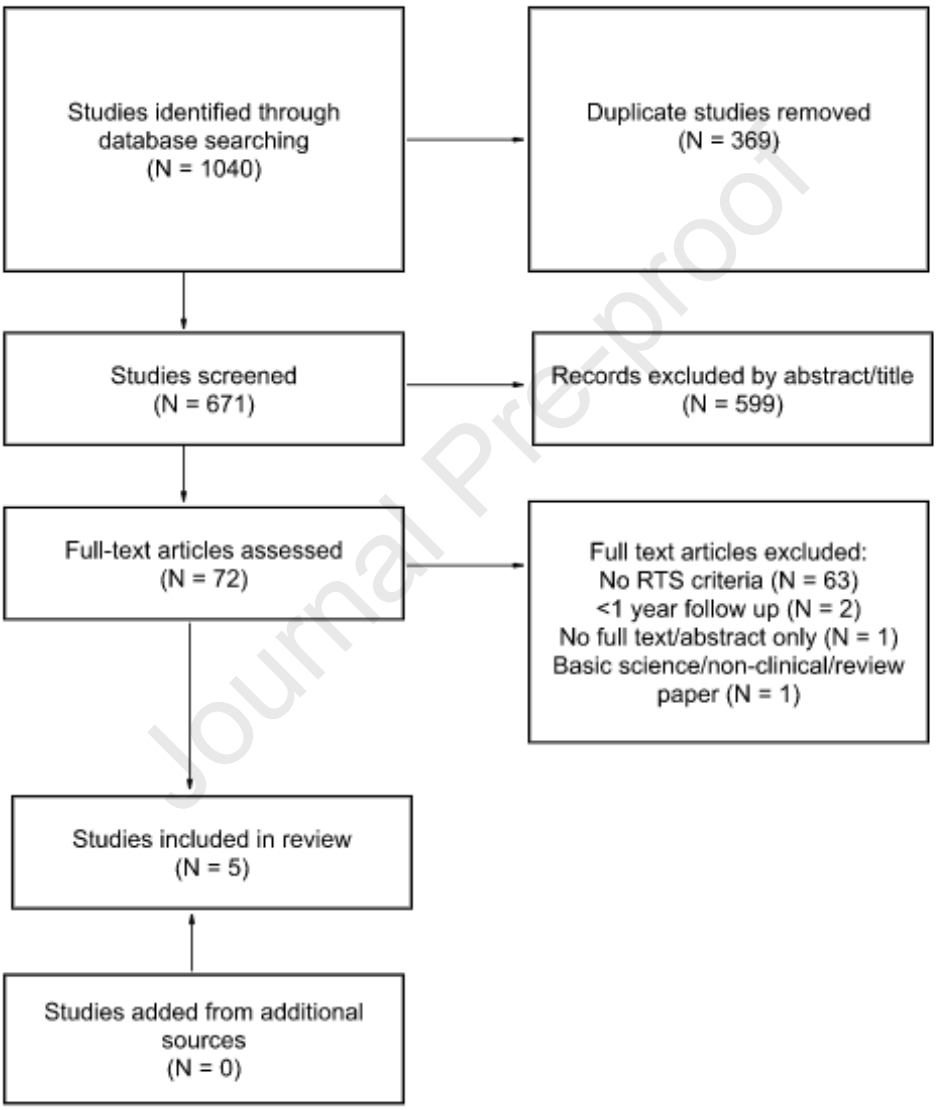
Figure 1

PRISMA (Preferred Reporting Items for Systematic Review and Meta-Analyses) flow diagram for the assessment strategy for included studies. RTS; return to sport



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